Executive Summary

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- We constructed the Cox PH baseline survival $S0(t)$ for the 2018–2020 cohort and validated checkpoints (2–7Y).
- A historical prior of 0.28 at 4Y was time-aligned across t using the ratio of baseline event accumulation from $S0(t)$.
- At 5Y, the aligned historical event probability (~ 0.456) matches the Cox baseline event probability ($1 - S0(5) \sim 0.455$).
- For a single 5Y probability, blend $p\_blend = 0.4 \cdot p\_hist\_adj(5Y) + 0.6 \cdot [1 - S0(5Y)] \approx 0.455$.
- Use $S(t|x) = S0(t)^{\exp(\beta^T x)}$ to compute individualized probabilities at any horizon t.
- Files: breslow_baseline_survival_2018_2020.csv and adjusted_historical_prior_aligned.csv.

Technical Note: Baseline Survival and Prior Alignment (2018–2020)

Purpose
This note documents methods to reproduce: (i) historical prior, (ii) Breslow baseline survival S0(t), and (iii) adjusted historical prior aligned to S0(t).

Data Inputs

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- Cohort: 2018–2020; Time: reporting_years; Event: end_success.
- Features: cleantech_flag; cost_quintile; province dummies (AB baseline); sector dummies (Energy baseline).
- Coefficients: cox_coefficients_with_references.csv; Outputs: two CSVs named above.

(i) Historical Prior

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Prior = mean(end_success) in cohort (~0.28). Not time-specific; empirical anchor.

(ii) Breslow Baseline Survival S0(t)

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Cox: $\eta = \beta^T x$; $r = \exp(\eta)$. $H0\hat{}(t) = \Sigma\, d(\tau)/\Sigma\, r_j$ over risk set; $S0(t) = \exp(-H0\hat{}(t))$. Baseline: non-cleantech; cost_quintile=0; AB; Energy.

(iii) Adjusted Historical Prior (Aligned)

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p_hist_adj(t) = 0.28 × [(1 − S0(t)) / (1 − S0(4))]. Saved with baseline_event_prob to aligned CSV.

Blending (Optional)

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p_blend(t) = w·p_hist_adj(t) + (1−w)·[1 − S0(t)], suggest w=0.4 historic, 0.6 Cox.

Notes & QC

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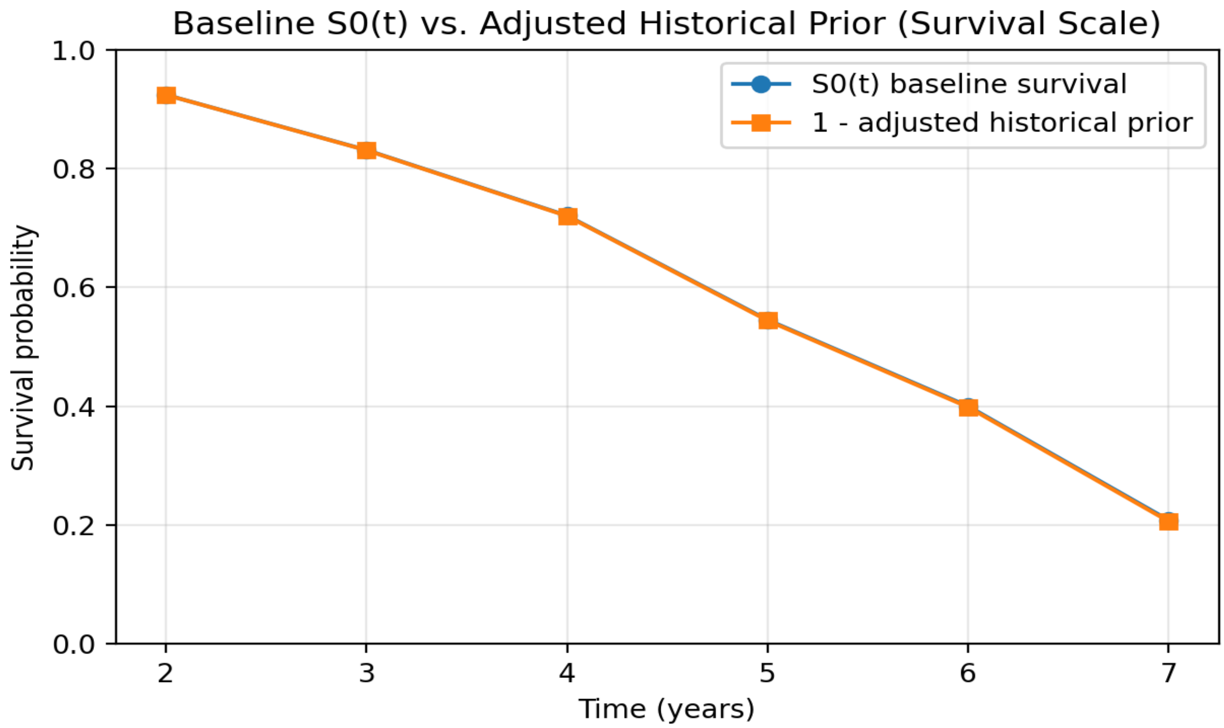
Numeric dtypes; integerized time; drop baseline dummies; align coefficient names; interpolate only for presentation.

Figure: Baseline S0(t) vs. Adjusted Historical Prior (Survival Scale)


Baseline S0(t) vs. Adjusted Historical Prior (Survival Scale)
S0(t) baseline survival
1 - adjusted historical prior
Survival probability
Time (years)
0.0
0.2
0.4
0.6
0.8
1.0
2
3
4
5
6
7